

Intertextuality in diachronic Greek

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Nikolaos Lavidas

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What AthDGC contributes

Nikolaos Lavidas, NKUA Athens Digital Glossa Chronos Research Network *Διαχρονία
Γλώσσας :Χρόνος*

HFRI Project No. 20577 · Greece 2.0 NRRP · Compute GRNET ARIS pa260305

1. The premise. Historical linguistics works on texts, not speakers.
2. What intertextuality means inside a diachronic-Greek setting.
3. Three modes: retelling, retranslation, inner-textual citation.
4. The hypothesis: parallel coexistent grammars in written contact (Lavidas 2021, Brill).
5. How AthDGC operationalises the hypothesis: one PROIEL schema across 3000 years.
6. Concrete platform, concrete numbers, three threads through the corpus.
7. What this offers a computational-intertextuality programme.

Nikolaos Lavidas (PI, NKUA) · **Kiki Nikiforidou** (NKUA) **Dag Haug** (Oslo, PROIEL Director) · **Leonid Kulikov** (Ghent) **Vassiliki Geka** (NKUA) · **Vassileios Symeonidis** (NKUA) · **Theodoros Michalareas** (NKUA) **Sofia Chionidi** (NKUA) · **Anastasia Tsiropina** (NKUA) · **Eleni Plakoutsi** (NKUA) · **Evangelos Argyropoulos** (NKUA)

Three universities (NKUA · Oslo · Ghent); eleven researchers; one platform. HFRI Project No. 20577 · Greece 2.0 NRRP.

1. The premise

Historical linguistics has **no native speakers**. It has only **texts**.

The only window onto Archaic, Classical, Koine, Late Antique, Byzantine, Late Byzantine, and Early Modern Greek is the written record. Every claim about the syntax of a period reduces to a claim about what its surviving texts permit and what they exclude.

2. Intertextuality in a diachronic-Greek setting

In a corpus that spans **3000 years on one language**, intertextuality is not a stylistic ornament. It is the **primary data-generating process**.

The same canon is **re-rendered** by every generation:

- the same Homeric verse is paraphrased by Byzantine schoolbooks;
- the same Pauline epistle is retranslated into Modern Greek every century;
- the same Septuagint formula is quoted, glossed, and inflected toward each period's spoken grammar.

3. Three modes

Retelling. Same language, later century. Same story, new register. (*Niketas Choniates retelling Homer in 12th c. Byzantine epitome.*)

Retranslation. Same canon, target language shifts. (*Hebrew Bible → Greek Septuagint → Latin Vulgate → Gothic Wulfila → OCS Marianus → Classical Armenian.*)

Inner-textual citation. Surface re-use of one text inside another. (*NT quoting LXX; Patristic homily quoting NT; Byzantine chronicle quoting Patristic.*)

4. The hypothesis

Under sustained **written language contact**, donor and recipient grammars do not displace one another. They **coexist** as distinct active systems inside the same writer.

Lavidas, N. 2021. *The Diachrony of Written Language Contact: A Contrastive Approach*. Brill's Studies in Historical Linguistics. Leiden, Boston: Brill.

5. Why “written” matters

Written transmission has its own physics:

- **prestige standards** (Atticism, Byzantine Hochsprache);
- **copying chains** with attested manuscript stemmata;
- **citation as authority** (citing scripture *as scripture*);
- **resistance to vernacular pressure** at literate registers.

Each of these leaves a different signature in the syntactic record.

6. One schema across 3000 years

PROIEL XML 2.0 is the file format that stores each sentence as a tree of word-by-word grammatical relations. Developed at Oslo for the early Indo-European languages; extended in AthDGC to all of Greek.

The same 26-relation inventory annotates:

- Homer (8th c. BCE)
- the New Testament (1st c. CE)
- Sphrantzes' Chronicle (15th c. CE)
- the Modern Greek standard (21st c. CE)

One schema; eight periods; queryable comparison.

The diachronic extension was first proposed by Lavidas and Haug (2012, Thessaloniki-Oslo PROIEL pilot on Sphrantzes' Chronicle). AthDGC v0.4 ships that 2012 idea, generalised to the full diachronic span.

7. Text Id

Every partition carries a **stable identity**:

athdgc.<author>.<work>.<src_lang>.<tgt_lang>.<translator>.v<revision>

Example:

athdgc.017.001.grc.000.000.v1 (Sphrantzes, Chronicle, Greek)

Every Text Id resolves to a row in a metadata register, to a source edition, and to a citable printed reference.

8. What is built (tools)

- An **annotation pipeline** that reads a Greek sentence and emits PROIEL XML 2.0 (the file format that stores each sentence as a tree of word-by-word grammatical relations, developed at Oslo).
- A **morphological tagger** (Stanford **Stanza**, in its PROIEL-trained version), which produces lemma + part-of-speech + features automatically.
- An **AI second-opinion review loop**: uncertain annotations are flagged and re-checked by a large language model, then approved by a human.
- A **live dashboard** that shows token counts and per-period progress in real time as the annotation runs.

All open source. MIT · Apache 2.0 · BSD.

9. What is built (corpus)

Eight diachronic periods, all on one PROIEL XML 2.0 spine:

Period	Status
Archaic, Classical, Hellenistic, Koine	partitions in flight
Late Antique, Byzantine	partitions in flight
Late Byzantine, Early Modern, Modern	partitions in flight

New Testament verse-level alignment (cross-alignment: machine-readable matching of corresponding sentences and words between texts in different languages):

- to **Latin** (Vulgate), **Gothic** (Wulfila), **Old Church Slavonic** (Marianus): aligned in v0.4;
- to **Classical Armenian**: workflow established, alignment in ingestion for v0.5.

10. What is built (platform)

- Public showcase: **athdgc.github.io**
- Corpus repository: **github.com/AthDGC/athdgc-corpus** (closed during v0.5 audit)
- Live PROIEL viewer: **dialing.enl.uoa.gr/proiel/**
- Launch report: **arXiv cs.CL** (published 11 Jun 2026)
- Permanent identifier: **Zenodo concept DOI 10.5281/zenodo.20439182**

CC BY 4.0 for the corpus. Citable per-version DOIs on each release.

11. Concrete numbers (today)

10.91 M tokens PROIEL-annotated **173** annotation batches complete **8** Greek periods covered **3** Indo-European parallels currently aligned at the New Testament

Dashboard snapshot: 19 June 2026 21:17 Athens. Latin, Gothic, Old Church Slavonic aligned in v0.4; Classical Armenian in ingestion for v0.5. v0.5 target 16 M tokens; v0.6 target 24 M.

12. Three threads through the corpus

Thread A. The Homeric chain. Iliad 1.1 in Archaic Greek → its Byzantine epitome (Tzetzes) → its 1955 modern Greek prose retelling (Kakridis-Kazantzakis).

Thread B. The New-Testament chain. John 1.1 in Koine → its Vulgate translation → its Gothic and Old Church Slavonic sisters.

Thread C. The Septuagint chain. Psalm 1.1 in Hebrew (target) ← LXX Greek → Vulgate → Modern Greek liturgy.

12a. A closer look at the Septuagint

The **Septuagint** (LXX) is the most consequential single text in the Greek chain:

- a **Koine-Greek translation of the Hebrew Bible**, made in stages from the 3rd c. BCE (Pentateuch) onward;
- the **Greek Bible of the early Church**: New Testament writers quote it, not the Hebrew;
- the **source of the first Christian translations** into Latin (Old Latin, then Vulgate for many books), Gothic (Wulfila), Old Church Slavonic (Cyril and Methodius), and Classical Armenian;
- still in use in the **Modern Greek liturgy** today.

12b. What LXX shows on a PROIEL spine

Two well-studied phenomena that AthDGC makes **queryable across periods**:

1. **Paratactic καί** as a calque of the Hebrew *waw*-consecutive: clause-initial καί + finite verb, instead of classical Greek subordinate-clause patterns;
2. **Closest-conjunct agreement**: number and gender agreement track the nearest conjunct under Hebrew influence (Lavidas 2019, *Questions and Answers in Linguistics* 5(2): 37-90).

Both are syntactic transfers from Hebrew. Both propagate outward, through Vulgate, Gothic, Old Church Slavonic.

12c. The Septuagint, a concrete example

Genesis 1:3-4 (the *fiat lux* sequence)

Heb. וַיֵּרָא אֱלֹהִים וְהָיָה אֵלֶּיךָ אֱלֹהִים וַיֵּרָא אֱלֹהִים וְהָיָה אֵלֶּיךָ אֱלֹהִים
וַיֵּרָא אֱלֹהִים וְהָיָה אֵלֶּיךָ אֱלֹהִים וַיֵּרָא אֱלֹהִים וְהָיָה אֵלֶּיךָ אֱלֹהִים

Tr. wa-yyō'mer 'ēlōhîm yəhî 'ôr wa-yhî 'ôr. wa-yyar' 'ēlōhîm 'et ha-'ôr kî tōb.

LXX καὶ εἶπεν ὁ θεός Γενηθήτω φῶς. καὶ ἐγένετο φῶς. καὶ εἶδεν ὁ θεὸς τὸ φῶς ὅτι καλόν.

Gloss and said the god let-become light. and became light. and saw the god the light that good.

Three Hebrew *wayyiqtol* forms (consecutive narrative tense) are calqued by three clause-initial καί + finite verb in the LXX. Classical Greek prose would subordinate (participle, μέν / δέ, ὅτε clause).

12d. Three thousand years, one schema

The eight periods of Greek on the AthDGC PROIEL spine:

13. What this offers the intertextuality programme

1. A test bed: 3000 years on one schema, queryable across periods.
2. Structural detection on top of surface detection: not “the same words were re-used”, but **“the same propositional content was re-encoded into a different syntax”**.
3. An empirical handle on the parallel-coexistent-grammars hypothesis: every retelling and retranslation is a controlled minimal pair.
4. Reusable infrastructure: the same pipeline can run on Hebrew, Sanskrit, Tibetan, with the relation inventory adjusted upstream.

14. Resources

- Public site: <https://athdgc.github.io>
- Corpus repository: <https://github.com/AthDGC/athdgc-corpus> (under v0.5 audit)
- Tools organisation: <https://github.com/AthDGC>
- Live PROIEL viewer: <https://dialing.enl.uoa.gr/proiel/>
- arXiv preprint: cs.CL (June 2026)
- Zenodo concept DOI: 10.5281/zenodo.20439182

Contact: nikolaos.lavidas@gmail.com · nlavidas@enl.uoa.gr

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